Gautam Buddha University

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Project Report

Disease Prediction Using Machine Learning

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Abstract

This project presents a machine learning approach to disease prediction based on binary symptom data. Using the publicly available Kaggle dataset by Kaushil268, which comprises 132 symptom features mapped to 41 disease classes, three classification models were trained and evaluated: Decision Tree, Random Forest, and Bernoulli Naive Bayes. The dataset is perfectly balanced with 120 training samples per disease and a pre-provided test set of 42 samples. Comprehensive exploratory data analysis was conducted to understand symptom distributions and disease-symptom relationships. All three models achieved very high accuracy, with Bernoulli Naive Bayes reaching 100% on the test set, and Decision Tree and Random Forest each achieving 97.62%. The results demonstrate that well-structured binary symptom data is highly amenable to classification-based disease diagnosis.

Keywords: Disease Prediction, Multi-class Classification, Random Forest, Bernoulli Naive Bayes, Decision Tree, Symptom Analysis

1. Introduction

Early and accurate disease diagnosis is one of the most critical steps in effective healthcare delivery. In clinical practice, physicians rely on a combination of patient-reported symptoms, test results, and medical history to arrive at a diagnosis. However, this process is time-consuming, prone to human error, and heavily dependent on the expertise of the practitioner available.

Machine learning offers a powerful alternative by learning diagnostic patterns from historical data. Given a structured set of symptoms, a trained classifier can predict the most probable disease — rapidly and consistently. This is especially useful in resource-limited settings where specialist access is restricted.

This project builds a symptom-based disease prediction system using three machine learning classifiers trained on a publicly available dataset of 4920 patient records covering 41 distinct diseases. The goal is to compare model performance, understand which symptoms are most diagnostically significant, and evaluate how well each algorithm handles the multi-class classification task.

The project was implemented in Python using the scikit-learn library, with Pandas and Matplotlib/Seaborn used for data manipulation and visualization respectively.

2. Dataset Description

2.1 Source

The dataset is sourced from Kaggle: *Disease Prediction Using Machine Learning* by Kaushil268.
URL: `https://www.kaggle.com/datasets/kaushil268/disease-prediction-using-machine-lear`

2.2 Structure

The dataset is provided as two separate CSV files:

Table 1: Dataset Summary

File	Rows	Columns
Training.csv	4920	133
Testing.csv	42	133

Of the 133 columns, 132 are binary symptom features (1 = symptom present, 0 = absent) and 1 is the target column `prognosis` containing the disease name.

2.3 Target Classes

The dataset covers 41 disease classes including Fungal infection, Allergy, GERD, Diabetes, Malaria, Dengue, Typhoid, Tuberculosis, Hepatitis variants (A through E), Common Cold, Pneumonia, Heart attack, Migraine, and others.

2.4 Class Balance

Each disease class has exactly 120 training samples, making this a perfectly balanced dataset with no class imbalance issues.

3. Data Preprocessing

The following preprocessing steps were applied before model training:

1. **Dropping extra column:** The training CSV contained an extra unnamed column (`Unnamed: 133`) absent from the test set. This was identified and dropped to align both files to 133 columns.
2. **Null value check:** Both training and test sets were verified to contain zero null values, confirming a clean dataset.
3. **Feature-target split:** The 132 symptom columns were extracted as the feature matrix X, and the `prognosis` column was assigned as the target vector y.
4. **Label encoding:** The target column contained string disease names. These were converted to integer class indices using scikit-learn's `LabelEncoder`.
5. **No feature scaling:** Since all features are binary (0 or 1), normalization and standardization were not required.
6. **Train-test split:** The Kaggle-provided split was used directly — `Training.csv` for training and `Testing.csv` for evaluation. No additional splitting was performed.

4. Exploratory Data Analysis

4.1 Disease Class Distribution

An analysis of the training set reveals that every disease class contains exactly 120 samples, confirming a perfectly balanced dataset. This is significant because it means no class will be over- or under-represented during training, and accuracy is a reliable evaluation metric without any need for re-sampling or class-weighting techniques.

4.2 Top 20 Most Frequent Symptoms

Examining symptom frequencies across all training records, *fatigue* and *vomiting* emerge as the most commonly reported symptoms, each occurring in nearly 2000 records out of 4920. Other frequently occurring symptoms include *high_fever*, *loss_of_appetite*, *nausea*, and *headache*. These are generic symptoms shared across many diseases, which reflects real-world diagnostic complexity — a single common symptom is rarely sufficient for diagnosis and must be considered in combination with others.

4.3 Total Active Symptoms per Disease

Aggregating active symptom counts per disease reveals a wide range of diagnostic complexity. *Common Cold* and *Tuberculosis* have the highest total symptom loads, indicating that these diseases present with a broader and more diverse set of symptoms. In contrast, *Fungal infection* and *Allergy* have the fewest active symptoms. This pattern reflects real medical complexity — systemic and respiratory diseases tend to involve more symptoms than localized skin conditions.

4.4 Disease-Symptom Associations

Analyzing the mean presence of the top 20 symptoms for each disease reveals several notable patterns:

- Hepatitis variants (A, B, C, D, E) share a strong association with *yellowish_skin* and *yellowing_of_eyes*, directly reflecting their shared hepatic pathology.
- Malaria and Dengue show high co-occurrence of *chills* and *high_fever*, consistent with their vector-borne fever profiles.
- Most diseases exhibit a sparse but distinct symptom signature, meaning each disease activates only a specific subset of the 132 features. This distinctiveness is the primary reason classifiers achieve very high accuracy on this dataset.

5. Models Used

Three classification models were trained and compared in this project.

5.1 Decision Tree

A Decision Tree recursively partitions the feature space by selecting the symptom that best splits the data at each node, according to a criterion such as Gini impurity or information gain. The result is a tree-structured model where each leaf node represents a disease class.

Decision Trees are highly interpretable — the path from root to leaf directly represents a diagnostic rule.

In this project, scikit-learn's `DecisionTreeClassifier` was used with `random_state=42` and default hyperparameters.

5.2 Random Forest

Random Forest is an ensemble learning method that constructs multiple Decision Trees during training and outputs the class predicted by the majority of trees. Each tree is trained on a bootstrapped sample of the data, and at each split only a random subset of features is considered. This introduces diversity among trees and reduces overfitting.

In this project, 100 trees were trained (`n_estimators=100`) using scikit-learn's `RandomForestClassifier` with `random_state=42`.

5.3 Bernoulli Naive Bayes

Bernoulli Naive Bayes is a probabilistic classifier based on Bayes' theorem with the assumption that features are binary and conditionally independent given the class. For a feature vector $\mathbf{x}$ and class c, the model computes:

$$P(c \mid \mathbf{x}) \propto P(c) \prod_{i=1}^{n} P(x_i \mid c)$$

where each $P(x_i \mid c)$ models the probability of symptom i being present or absent given disease c. Since all features in this dataset are binary (0/1), Bernoulli NB is theoretically the most natural fit among the three models.

6. Results and Evaluation

6.1 Accuracy Comparison

Table 2 summarizes the test set accuracy of all three models. Bernoulli NB achieves perfect accuracy, while Decision Tree and Random Forest both misclassify exactly one sample out of 42.

Table 2: Model Accuracy on Test Set

Model	**Accuracy**
Decision Tree	97.62%
Random Forest	97.62%
Bernoulli NB	**100.00%**

6.2 Confusion Matrices

Confusion matrix analysis across all three models shows near-perfect diagonals, confirming that misclassifications are rare and isolated. The vast majority of disease classes are predicted with both precision and recall equal to 1.00.

6.3 Misclassification Analysis

The one misclassified sample in both Decision Tree and Random Forest involves *Fungal infection*, which was predicted as *Hepatitis C* (Decision Tree) and *Impetigo* (Random Forest). This is medically plausible — Fungal infection shares overlapping skin-related symptoms (itching, skin rash) with both Hepatitis C and Impetigo, making it the most challenging boundary in this dataset.

Bernoulli NB correctly classified all 42 samples, including both Fungal infection test instances, owing to its strong theoretical alignment with binary feature distributions.

7. Discussion

7.1 Why Bernoulli NB Outperforms

Bernoulli Naive Bayes achieves 100% accuracy on this particular test set because its probabilistic structure is ideally suited to binary presence/absence data. Each symptom contributes an independent probability factor to the final class score, and with 132 binary features, the joint probability products create very sharp separation between disease classes. Since the dataset is also perfectly clean (no noise, no missing values), the independence assumption of NB does not introduce significant error.

7.2 Real-World Considerations

While all three models perform exceptionally well on this dataset, the controlled nature of the data must be acknowledged. In real clinical settings, symptom data is often noisy, incomplete, and subjective. Under such conditions, Random Forest would be the recommended model due to its robustness to noise, resistance to overfitting, and ability to handle correlated features — properties that Decision Trees and Naive Bayes lack.

7.3 Dataset Limitations

The test set contains only 42 samples (one per disease), which limits the statistical reliability of accuracy as a metric. A more rigorous evaluation would use cross-validation on the training set to obtain averaged performance estimates with confidence intervals.

8. Conclusion

This project successfully built and evaluated a multi-class disease prediction system using three machine learning classifiers trained on 132 binary symptom features. Key findings are summarized below:

- The dataset is perfectly balanced with 120 samples per disease, and contains no null values, making it an ideal classification benchmark.
- Exploratory data analysis revealed that generic symptoms like *fatigue* and *vomiting* are the most frequent, while diseases like Hepatitis variants share overlapping symptom signatures.
- All three models achieved very high accuracy — Bernoulli NB reached 100%, and Decision Tree and Random Forest both achieved 97.62%.
- The only misclassification involved *Fungal infection*, which shares skin-related symptoms with other diseases.
- For real-world deployment, Random Forest is the most robust choice due to its ensemble nature and noise tolerance.

This project demonstrates that machine learning can serve as a strong first-line diagnostic tool when structured symptom data is available, with potential applications in telemedicine, automated triage, and clinical decision support systems.

References

1. Kaushil268. *Disease Prediction Using Machine Learning*. Kaggle Dataset, 2020. https://www.kaggle.com/datasets/kaushil268/disease-prediction-using-machine-lear

2. Breiman, L. (2001). Random Forests. *Machine Learning*, 45, 5–32.

3. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

4. Mitchell, T. M. (1997). *Machine Learning*. McGraw-Hill.

5. Rish, I. (2001). An empirical study of the Naive Bayes classifier. *IJCAI Workshop on Empirical Methods in AI*.